

Local Results Summary — Uniform Sweep 0.9 / 0.8 / 0.7 (filled)

Filled from on-disk step-25 evals (coverage extract). **Local** = our measured score for baseline / +Uniform 0.9 / 0.8 / 0.7; **Paper** = BRV-published (baseline & uniform-0.9 only, for the 3 BRV models); **Gap** = Local – Paper; each Δ = (+Uniform α) – Baseline. — means no score on disk yet (eval still in flight). Table 2: previously-blocked models, now resolved.

All values, row-by-row (consolidated reference) — filled

Δ Local cells colored by the perception/reasoning rule, thresholded at each benchmark's Real Δ (pp) noise floor (yellow = favorable ≥ floor, red = unfavorable ≥ floor; orange/pink = below the floor, i.e. within noise). Red Gap cells = |Local – Paper| ≥ that same floor. MathVerse-Overall is the mean of its five split rows (only when all five are present). **Real Δ (pp)** = smallest baseline→config change that beats the 95% sampling-noise floor for that benchmark's item count n ($1.96 \cdot \sqrt{(0.5/n)}$); any Δ below it is statistical noise. Per-benchmark noise and references: final table.

	Yellow — favorable Δ that beats the benchmark's Real Δ (pp) noise floor (a real improvement).
	Red — unfavorable Δ that beats the floor (a real regression); also a Gap cell where Local – Paper ≥ the floor.
	Orange — Δ below the floor on a perception benchmark (within sampling noise).
	Pink — Δ below the floor on a reasoning / both benchmark (within sampling noise).

Real Δ (pp) noise floor and source per benchmark (the threshold used for cell coloring):

Benchmark	Real Δ (pp)	Reference	Benchmark	Real Δ (pp)	Reference
MathVista-Overall ↑	≥4.4	MathVista, arXiv:2310.02255	MathVista-General (NEW) ↑	≥6.5	MathVista, arXiv:2310.02255
MathVista-Math (NEW) ↑	≥6.0	MathVista, arXiv:2310.02255	MathVerse-Overall ↑	≥4.9	MathVerse, arXiv:2403.14624
MathVerse-T-D ↑	≥4.9	MathVerse, arXiv:2403.14624	MathVerse-T-L ↑	≥4.9	MathVerse, arXiv:2403.14624
MathVerse-V-I ↑	≥4.9	MathVerse, arXiv:2403.14624	MathVerse-V-D ↑	≥4.9	MathVerse, arXiv:2403.14624
MathVerse-V-O ↑	≥4.9	MathVerse, arXiv:2403.14624	MathVision ↑	≥7.9	MATH-Vision, arXiv:2402.14804
MMStar-Overall ↑	≥3.6	MMStar, arXiv:2403.20330	MMStar-Math ↑	≥8.8	MMStar, arXiv:2403.20330
DynaMath-Average ↑	≥2.0	DynaMath, arXiv:2411.00836 (rep. var. ~1-2pp)	DynaMath-Worst (NEW) ↑	≥6.2	DynaMath, arXiv:2411.00836 (rep. var. ~1-2pp)

						Baseline		+Uniform 0.9			+Uniform 0.8		Uniform 0.7	Δ 0.9 vs base		Δ 0.8	Δ 0.7
Model	Benchmark	n (Local)	n (BRV)	Type	Real Δ (pp)	Local	Paper	Gap	Local	Paper	Gap	Local	Local	Local	Paper	Local	Local
LLaVA-Next-LLaMA3-8B (Dart-Prop)	MathVista-Overall ↑	1000	1000	both	≥4.4	38.10	37.40	+0.70	38.00	38.00	+0.00	37.60	36.80	-0.10	+0.60	-0.50	-1.30
	MathVista-General (NEW) ↑	460	460	perception	≥6.5	38.95	51.70	-12.75	38.68	48.70	-10.02	38.41	37.79	-0.27	-3.00	-0.54	-1.16
	MathVista-Math (NEW) ↑	540	540	reasoning	≥6.0	25.74	25.40	+0.34	28.52	28.90	-0.38	29.81	30.00	+2.78	+3.50	+4.07	+4.26
	MathVerse-Overall ↑	788*	788*	reasoning	≥4.9	18.55	20.10	-1.55	19.03	23.70	-4.67	20.18	18.12	+0.48	+3.60	+1.62	-0.43
	MathVerse-T-D ↑	788	788	reasoning	≥4.9	22.59	25.90	-3.31	23.60	30.70	-7.10	26.40	23.73	+1.01	+4.80	+3.81	+1.14
	MathVerse-T-L ↑	788	788	reasoning	≥4.9	19.80	20.80	-1.00	20.05	24.80	-4.75	21.45	19.54	+0.25	+4.00	+1.65	-0.26
	MathVerse-V-I ↑	788	788	reasoning	≥4.9	19.80	21.10	-1.30	20.05	25.50	-5.45	21.57	18.91	+0.25	+4.40	+1.77	-0.89
	MathVerse-V-D ↑	788	788	reasoning	≥4.9	15.61	16.50	-0.89	15.86	19.80	-3.94	15.99	15.74	+0.25	+3.30	+0.38	+0.13
	MathVerse-V-O ↑	788	788	reasoning	≥4.9	14.97	16.00	-1.03	15.61	17.40	-1.79	15.48	12.69	+0.64	+1.40	+0.51	-2.28
	MathVision ↑	304	304	reasoning	≥7.9	14.80	13.80	+1.00	14.80	14.80	+0.00	13.49	13.82	+0.00	+1.00	-1.32	-0.99
	MMStar-Overall ↑	1500	1500	both	≥3.6	43.33	43.80	-0.47	43.60	43.60	+0.00	43.53	42.53	+0.27	-0.20	+0.20	-0.80
	MMStar-Math ↑	250	250	reasoning	≥8.8	29.60	30.00	-0.40	33.60	33.60	+0.00	30.40	34.00	+4.00	+3.60	+0.80	+4.40
	DynaMath-Average ↑	5010	5010	reasoning	≥2.0	22.61	22.70	-0.09	24.23	24.50	-0.27	24.19	24.53	+1.62	+1.80	+1.58	+1.92
Idefics2-8B (MAmmoTH-1)	DynaMath-Worst (NEW) ↑	501	501	reasoning	≥6.2	3.39	—	—	4.39	—	—	3.59	5.59	+1.00	—	+0.20	+2.20
	MathVista-Overall ↑	1000	1000	both	≥4.4	52.00	51.80	+0.20	53.10	53.00	+0.10	51.60	46.30	+1.10	+1.20	-0.40	-5.70
	MathVista-General (NEW) ↑	460	460	perception	≥6.5	51.04	57.00	-5.96	52.09	58.50	-6.41	50.43	45.35	+1.06	+1.50	-0.61	-5.69

						Baseline			+Uniform 0.9			+Uniform 0.8		+Uniform 0.7		Δ 0.9 vs base		Δ 0.8	Δ 0.7
Model	Benchmark	n (Local)	n (BRV)	Type	Real Δ (pp)	Local	Paper	Gap	Local	Paper	Gap	Local	Local	Local	Paper	Local	Local		
	MathVista-Math (NEW) ↑	540	540	reasoning	≥6.0	48.33	47.40	+0.93	48.52	48.30	+0.22	47.41	41.48	+0.19	+0.90	-0.93	-6.85		
	MathVerse-Overall ↑	788*	788*	reasoning	≥4.9	18.17	19.40	-1.23	18.78	20.40	-1.62	18.73	19.47	+0.61	+1.00	+0.56	+1.29		
	MathVerse-T-D ↑	788	788	reasoning	≥4.9	21.83	24.40	-2.57	22.84	26.00	-3.16	21.83	25.25	+1.01	+1.60	+0.00	+3.42		
	MathVerse-T-L ↑	788	788	reasoning	≥4.9	20.18	21.30	-1.12	20.69	22.50	-1.81	21.32	20.94	+0.51	+1.20	+1.14	+0.76		
	MathVerse-V-I ↑	788	788	reasoning	≥4.9	19.54	20.70	-1.16	19.67	21.30	-1.63	20.30	19.67	+0.13	+0.60	+0.76	+0.13		
	MathVerse-V-D ↑	788	788	reasoning	≥4.9	19.04	19.70	-0.66	18.78	19.80	-1.02	18.40	18.15	-0.26	+0.10	-0.64	-0.89		
	MathVerse-V-O ↑	788	788	reasoning	≥4.9	10.28	11.00	-0.72	11.93	12.10	-0.17	11.80	13.32	+1.65	+1.10	+1.52	+3.04		
	MathVision ↑	304	304	reasoning	≥7.9	16.78	17.10	-0.32	17.76	16.80	+0.96	18.42	18.75	+0.99	-0.30	+1.64	+1.97		
	MMStar-Overall ↑	1500	1500	both	≥3.6	49.33	49.50	-0.17	48.40	48.30	+0.10	47.07	44.80	-0.93	-1.20	-2.27	-4.53		
	MMStar-Math ↑	250	250	reasoning	≥8.8	39.60	39.60	+0.00	40.00	40.80	-0.80	41.20	38.80	+0.40	+1.20	+1.60	-0.80		
	DynaMath-Average ↑	5010	5010	reasoning	≥2.0	21.80	21.80	-0.00	23.01	23.20	-0.19	23.71	24.09	+1.22	+1.40	+1.92	+2.30		
	DynaMath-Worst (NEW) ↑	501	501	reasoning	≥6.2	3.19	—	—	4.79	—	—	4.79	4.99	+1.60	—	+1.60	+1.80		
Qwen2-VL-7B-Instruct (Qwen2-Math-7B)	MathVista-Overall ↑	1000	1000	both	≥4.4	61.10	61.20	-0.10	58.40	60.20	-1.80	48.80	38.00	-2.70	-1.00	-12.30	-23.10		
	MathVista-General (NEW) ↑	460	460	perception	≥6.5	60.33	69.60	-9.27	58.02	68.00	-9.98	48.72	38.51	-2.32	-1.60	-11.61	-21.83		
	MathVista-Math (NEW) ↑	540	540	reasoning	≥6.0	53.70	54.10	-0.40	51.48	53.50	-2.02	44.26	31.11	-2.22	-0.60	-9.44	-22.59		
	MathVerse-Overall ↑	788*	788*	reasoning	≥4.9	30.68	31.80	-1.12	30.31	31.90	-1.59	27.46	22.94	-0.38	+0.10	-3.22	-7.74		
	MathVerse-T-D ↑	788	788	reasoning	≥4.9	34.77	35.90	-1.13	33.63	37.10	-3.47	30.33	25.63	-1.14	+1.20	-4.44	-9.14		
	MathVerse-T-L ↑	788	788	reasoning	≥4.9	30.58	31.40	-0.82	30.33	31.70	-1.37	26.90	23.73	-0.25	+0.30	-3.68	-6.85		
	MathVerse-V-I ↑	788	788	reasoning	≥4.9	30.71	31.50	-0.79	29.70	31.50	-1.80	28.05	23.98	-1.01	+0.00	-2.66	-6.73		
	MathVerse-V-D ↑	788	788	reasoning	≥4.9	32.23	33.10	-0.87	30.84	32.50	-1.66	27.16	24.24	-1.39	-0.60	-5.07	-7.99		
	MathVerse-V-O ↑	788	788	reasoning	≥4.9	25.13	26.90	-1.77	27.03	26.70	+0.33	24.87	17.13	+1.90	-0.20	-0.26	-8.00		
	MathVision ↑	304	304	reasoning	≥7.9	20.72	21.10	-0.38	19.08	21.70	-2.62	17.76	16.12	-1.64	+0.60	-2.96	-4.61		
	MMStar-Overall ↑	1500	1500	both	≥3.6	59.60	59.90	-0.30	56.20	59.50	-3.30	46.20	37.53	-3.40	-0.40	-13.40	-22.07		
	MMStar-Math ↑	250	250	reasoning	≥8.8	58.40	59.20	-0.80	54.40	58.40	-4.00	40.40	34.00	-4.00	-0.80	-18.00	-24.40		
	DynaMath-Average ↑	5010	5010	reasoning	≥2.0	33.55	34.40	-0.85	32.52	35.00	-2.48	22.22	16.55	-1.04	+0.60	-11.34	-17.01		
	DynaMath-Worst (NEW) ↑	501	501	reasoning	≥6.2	10.38	—	—	10.58	—	—	5.39	2.20	+0.20	—	-4.99	-8.18		
LLaVA-1.5-7B (MetaMath-7B-V1.0)	MathVista-Overall ↑	1000	1000	both	≥4.4	24.10	—	—	23.70	—	—	24.50	25.10	-0.40	—	+0.40	+1.00		
	MathVista-General (NEW) ↑	460	460	perception	≥6.5	24.21	—	—	23.63	—	—	24.51	25.20	-0.58	—	+0.31	+0.99		
	MathVista-Math (NEW) ↑	540	540	reasoning	≥6.0	17.04	—	—	18.33	—	—	18.52	19.26	+1.30	—	+1.48	+2.22		
	MathVerse-Overall ↑	788*	788*	reasoning	≥4.9	15.03	—	—	14.90	—	—	15.05	15.43	-0.13	—	+0.02	+0.41		
	MathVerse-T-D ↑	788	788	reasoning	≥4.9	15.61	—	—	14.85	—	—	15.86	16.88	-0.76	—	+0.25	+1.27		
	MathVerse-T-L ↑	788	788	reasoning	≥4.9	14.59	—	—	14.59	—	—	14.34	14.72	+0.00	—	-0.25	+0.13		
	MathVerse-V-I ↑	788	788	reasoning	≥4.9	14.85	—	—	14.72	—	—	14.34	14.34	-0.13	—	-0.51	-0.51		
	MathVerse-V-D ↑	788	788	reasoning	≥4.9	15.36	—	—	15.10	—	—	15.48	15.86	-0.26	—	+0.12	+0.50		
	MathVerse-V-O ↑	788	788	reasoning	≥4.9	14.72	—	—	15.23	—	—	15.23	15.36	+0.51	—	+0.51	+0.64		
	MathVision ↑	304	304	reasoning	≥7.9	15.79	—	—	17.11	—	—	18.09	18.42	+1.32	—	+2.30	+2.63		

						Baseline			+Uniform 0.9			+Uniform 0.8		+Uniform 0.7		Δ 0.9 vs base		Δ 0.8	Δ 0.7
Model	Benchmark	n (Local)	n (BRV)	Type	Real Δ (pp)	Local	Paper	Gap	Local	Paper	Gap	Local	Local	Local	Paper	Local	Local		
	MMStar-Overall ↑	1500	1500	both	≥3.6	31.07	—	—	32.73	—	—	34.20	33.60	+1.67	—	+3.13	+2.53		
	MMStar-Math ↑	250	250	reasoning	≥8.8	22.40	—	—	27.60	—	—	28.80	28.80	+5.20	—	+6.40	+6.40		
	DynaMath-Average ↑	5010	5010	reasoning	≥2.0	12.73	—	—	11.86	—	—	11.46	10.92	-0.88	—	-1.28	-1.82		
	DynaMath-Worst (NEW) ↑	501	501	reasoning	≥6.2	1.20	—	—	1.80	—	—	1.60	1.00	+0.60	—	+0.40	-0.20		
LLaVA-Next-Mistral-7B (Dart-Math-Mistral-Uniform)	MathVista-Overall ↑	1000	1000	both	≥4.4	35.40	—	—	36.00	—	—	33.80	33.50	+0.60	—	-1.60	-1.90		
	MathVista-General (NEW) ↑	460	460	perception	≥6.5	35.96	—	—	36.74	—	—	34.59	34.29	+0.78	—	-1.36	-1.67		
	MathVista-Math (NEW) ↑	540	540	reasoning	≥6.0	26.11	—	—	26.30	—	—	23.89	25.00	+0.19	—	-2.22	-1.11		
	MathVerse-Overall ↑	788*	788*	reasoning	≥4.9	18.27	—	—	17.92	—	—	18.63	18.93	-0.36	—	+0.35	+0.66		
	MathVerse-T-D ↑	788	788	reasoning	≥4.9	21.19	—	—	21.07	—	—	22.59	24.11	-0.12	—	+1.40	+2.92		
	MathVerse-T-L ↑	788	788	reasoning	≥4.9	17.89	—	—	18.27	—	—	18.65	19.92	+0.38	—	+0.76	+2.03		
	MathVerse-V-I ↑	788	788	reasoning	≥4.9	19.16	—	—	19.16	—	—	18.78	18.02	+0.00	—	-0.38	-1.14		
	MathVerse-V-D ↑	788	788	reasoning	≥4.9	17.01	—	—	16.75	—	—	16.75	15.99	-0.26	—	-0.26	-1.02		
	MathVerse-V-O ↑	788	788	reasoning	≥4.9	16.12	—	—	14.34	—	—	16.37	16.62	-1.78	—	+0.25	+0.50		
	MathVision ↑	304	304	reasoning	≥7.9	11.18	—	—	13.82	—	—	15.46	15.79	+2.63	—	+4.28	+4.61		
	MMStar-Overall ↑	1500	1500	both	≥3.6	39.33	—	—	38.40	—	—	36.67	36.53	-0.93	—	-2.67	-2.80		
	MMStar-Math ↑	250	250	reasoning	≥8.8	27.20	—	—	26.80	—	—	26.80	25.60	-0.40	—	-0.40	-1.60		
	DynaMath-Average ↑	5010	5010	reasoning	≥2.0	17.78	—	—	19.26	—	—	22.48	22.50	+1.48	—	+4.69	+4.71		
DynaMath-Worst (NEW) ↑	501	501	reasoning	≥6.2	2.20	—	—	2.40	—	—	4.19	4.19	+0.20	—	+2.00	+2.00			
LLaVA-OneVision-7B (Qwen2-Math-7B)	MathVista-Overall ↑	1000	1000	both	≥4.4	—	—	—	60.00	—	—	54.20	43.00	—	—	—	—		
	MathVista-General (NEW) ↑	460	460	perception	≥6.5	—	—	—	59.31	—	—	54.03	43.65	—	—	—	—		
	MathVista-Math (NEW) ↑	540	540	reasoning	≥6.0	—	—	—	61.48	—	—	55.74	41.11	—	—	—	—		
	MathVerse-Overall ↑	788*	788*	reasoning	≥4.9	—	—	—	30.36	—	—	28.78	24.82	—	—	—	—		
	MathVerse-T-D ↑	788	788	reasoning	≥4.9	—	—	—	38.71	—	—	36.42	31.60	—	—	—	—		
	MathVerse-T-L ↑	788	788	reasoning	≥4.9	—	—	—	33.63	—	—	31.35	27.03	—	—	—	—		
	MathVerse-V-I ↑	788	788	reasoning	≥4.9	—	—	—	31.73	—	—	30.08	24.62	—	—	—	—		
	MathVerse-V-D ↑	788	788	reasoning	≥4.9	—	—	—	29.06	—	—	28.05	24.11	—	—	—	—		
	MathVerse-V-O ↑	788	788	reasoning	≥4.9	17.64	—	—	18.65	—	—	18.02	16.75	+1.01	—	+0.38	-0.89		
	MathVision ↑	304	304	reasoning	≥7.9	18.42	—	—	17.43	—	—	17.11	13.16	-0.99	—	-1.32	-5.26		
	MMStar-Overall ↑	1500	1500	both	≥3.6	61.93	—	—	60.20	—	—	54.87	37.87	-1.73	—	-7.07	-24.07		
	MMStar-Math ↑	250	250	reasoning	≥8.8	58.80	—	—	57.20	—	—	53.20	38.80	-1.60	—	-5.60	-20.00		
	DynaMath-Average ↑	5010	5010	reasoning	≥2.0	19.66	—	—	20.72	—	—	32.12	28.04	+1.06	—	+12.46	+8.38		
DynaMath-Worst (NEW) ↑	501	501	reasoning	≥6.2	4.99	—	—	5.59	—	—	9.58	5.79	+0.60	—	+4.59	+0.80			
InternVL2.5-8B (InternLM2-Math-Plus-7B)	MathVista-Overall ↑	1000	1000	both	≥4.4	65.00	—	—	50.50	—	—	36.80	31.40	-14.50	—	-28.20	-33.60		
	MathVista-General (NEW) ↑	460	460	perception	≥6.5	64.59	—	—	51.07	—	—	37.52	31.56	-13.52	—	-27.07	-33.03		
	MathVista-Math (NEW) ↑	540	540	reasoning	≥6.0	61.48	—	—	40.19	—	—	26.30	27.41	-21.30	—	-35.19	-34.07		

						Baseline			+Uniform 0.9			+Uniform 0.8		+Uniform 0.7		Δ 0.9 vs base		Δ 0.8	Δ 0.7
Model	Benchmark	n (Local)	n (BRV)	Type	Real Δ (pp)	Local	Paper	Gap	Local	Paper	Gap	Local	Local	Local	Paper	Local	Local		
	MathVerse-Overall ↑	788*	788*	reasoning	≥4.9	34.67	—	—	27.36	—	—	14.47	1.62	-7.31	—	-20.20	-33.05		
	MathVerse-T-D ↑	788	788	reasoning	≥4.9	42.89	—	—	31.60	—	—	16.12	1.40	-11.29	—	-26.77	-41.49		
	MathVerse-T-L ↑	788	788	reasoning	≥4.9	38.58	—	—	29.06	—	—	14.97	1.40	-9.52	—	-23.61	-37.18		
	MathVerse-V-I ↑	788	788	reasoning	≥4.9	37.31	—	—	30.20	—	—	15.74	1.14	-7.11	—	-21.57	-36.17		
	MathVerse-V-D ↑	788	788	reasoning	≥4.9	33.25	—	—	26.14	—	—	14.21	1.52	-7.11	—	-19.04	-31.73		
	MathVerse-V-O ↑	788	788	reasoning	≥4.9	21.32	—	—	19.80	—	—	11.29	2.66	-1.52	—	-10.03	-18.66		
	MathVision ↑	304	304	reasoning	≥7.9	23.68	—	—	10.86	—	—	9.87	9.21	-12.83	—	-13.82	-14.47		
	MMStar-Overall ↑	1500	1500	both	≥3.6	62.73	—	—	61.13	—	—	57.47	20.67	-1.60	—	-5.27	-42.07		
	MMStar-Math ↑	250	250	reasoning	≥8.8	68.00	—	—	69.60	—	—	62.40	20.80	+1.60	—	-5.60	-47.20		
	DynaMath-Average ↑	5010	5010	reasoning	≥2.0	32.22	—	—	29.82	—	—	27.33	17.29	-2.40	—	-4.89	-14.93		
	DynaMath-Worst (NEW) ↑	501	501	reasoning	≥6.2	9.38	—	—	9.38	—	—	8.58	3.59	+0.00	—	-0.80	-5.79		
Qwen2.5-VL-7B (Qwen2.5-Math-7B)	MathVista-Overall ↑	1000	1000	both	≥4.4	68.90	—	—	63.40	—	—	37.80	28.50	-5.50	—	-31.10	-40.40		
	MathVista-General (NEW) ↑	460	460	perception	≥6.5	68.20	—	—	62.96	—	—	38.71	28.36	-5.24	—	-29.49	-39.84		
	MathVista-Math (NEW) ↑	540	540	reasoning	≥6.0	67.04	—	—	59.26	—	—	33.33	27.78	-7.78	—	-33.70	-39.26		
	MathVerse-Overall ↑	788*	788*	reasoning	≥4.9	44.01	—	—	37.84	—	—	21.50	17.05	-6.17	—	-22.52	-26.96		
	MathVerse-T-D ↑	788	788	reasoning	≥4.9	53.43	—	—	45.30	—	—	25.25	17.64	-8.13	—	-28.18	-35.79		
	MathVerse-T-L ↑	788	788	reasoning	≥4.9	45.18	—	—	37.94	—	—	21.45	18.65	-7.24	—	-23.73	-26.53		
	MathVerse-V-I ↑	788	788	reasoning	≥4.9	40.61	—	—	35.66	—	—	19.92	17.39	-4.95	—	-20.69	-23.22		
	MathVerse-V-D ↑	788	788	reasoning	≥4.9	40.10	—	—	36.04	—	—	21.95	16.62	-4.06	—	-18.15	-23.48		
	MathVerse-V-O ↑	788	788	reasoning	≥4.9	40.74	—	—	34.26	—	—	18.91	14.97	-6.48	—	-21.83	-25.77		
	MathVision ↑	304	304	reasoning	≥7.9	26.32	—	—	24.67	—	—	20.72	11.51	-1.64	—	-5.59	-14.80		
	MMStar-Overall ↑	1500	1500	both	≥3.6	61.67	—	—	60.27	—	—	41.93	30.80	-1.40	—	-19.73	-30.87		
	MMStar-Math ↑	250	250	reasoning	≥8.8	61.20	—	—	60.80	—	—	35.60	27.60	-0.40	—	-25.60	-33.60		
	DynaMath-Average ↑	5010	5010	reasoning	≥2.0	53.27	—	—	51.38	—	—	30.24	—	-1.90	—	-23.03	—		
	DynaMath-Worst (NEW) ↑	501	501	reasoning	≥6.2	22.36	—	—	20.16	—	—	6.99	—	-2.20	—	-15.37	—		
Qwen2.5-VL-3B (Qwen2.5-3B-DAPO-math-reasoning)	MathVista-Overall ↑	1000	1000	both	≥4.4	61.80	—	—	58.50	—	—	58.90	58.40	-3.30	—	-2.90	-3.40		
	MathVista-General (NEW) ↑	460	460	perception	≥6.5	61.46	—	—	58.26	—	—	58.63	57.75	-3.20	—	-2.83	-3.71		
	MathVista-Math (NEW) ↑	540	540	reasoning	≥6.0	57.78	—	—	52.41	—	—	52.78	54.44	-5.37	—	-5.00	-3.33		
	MathVerse-Overall ↑	788*	788*	reasoning	≥4.9	34.32	—	—	33.40	—	—	32.69	30.84	-0.92	—	-1.63	-3.48		
	MathVerse-T-D ↑	788	788	reasoning	≥4.9	41.75	—	—	41.62	—	—	39.47	37.94	-0.13	—	-2.28	-3.81		
	MathVerse-T-L ↑	788	788	reasoning	≥4.9	35.03	—	—	33.25	—	—	32.74	29.95	-1.78	—	-2.29	-5.08		
	MathVerse-V-I ↑	788	788	reasoning	≥4.9	32.11	—	—	31.85	—	—	30.46	30.96	-0.26	—	-1.65	-1.15		
	MathVerse-V-D ↑	788	788	reasoning	≥4.9	33.12	—	—	29.31	—	—	31.47	27.28	-3.81	—	-1.65	-5.84		
	MathVerse-V-O ↑	788	788	reasoning	≥4.9	29.57	—	—	30.96	—	—	29.31	28.05	+1.39	—	-0.26	-1.52		
	MathVision ↑	304	304	reasoning	≥7.9	23.36	—	—	27.30	—	—	23.68	21.38	+3.95	—	+0.33	-1.97		

						Baseline	+Uniform 0.9			+Uniform 0.8			+Uniform 0.7		Δ 0.9 vs base	Δ 0.8	Δ 0.7
Model	Benchmark	n (Local)	n (BRV)	Type	Real Δ (pp)	Local	Paper	Gap	Local	Paper	Gap	Local	Local	Local	Paper	Local	Local
	MMStar-Overall ↑	1500	1500	both	≥3.6	53.93	—	—	53.27	—	—	47.33	43.53	-0.67	—	-6.60	-10.40
	MMStar-Math ↑	250	250	reasoning	≥8.8	58.00	—	—	52.40	—	—	48.80	56.00	-5.60	—	-9.20	-2.00
	DynaMath-Average ↑	5010	5010	reasoning	≥2.0	42.26	—	—	42.02	—	—	41.10	39.88	-0.24	—	-1.16	-2.38
	DynaMath-Worst (NEW) ↑	501	501	reasoning	≥6.2	12.77	—	—	11.98	—	—	13.77	13.37	-0.80	—	+1.00	+0.60

Masked alpha sweep — PMBT gate_up (pmbt_t*_v1.0_m1.0_mlponly)

Masked task-arithmetic on text-classified MLP neurons (v=m=1.0, mlp-only). Local = measured score; Δ = vs the model's Baseline, colored at each benchmark's Real Δ (pp) noise floor (yellow = favorable ≥ floor, red = unfavorable ≥ floor; orange/pink = below the floor). — = no score on disk. MathVerse-Overall = mean of 5 splits. Real Δ (pp) = min change beating the 95% noise floor (see final-table references); treat any Δ below it as noise.

				Base	α=0.3		α=0.4		α=0.5		α=0.6		α=0.7		α=0.8		α=0.9	
Model	Benchmark	Type	Real Δ (pp)	Local	Local	Δ	Local	Δ	Local	Δ	Local	Δ	Local	Δ	Local	Δ	Local	Δ
LLaVA-Next-LLaMA3-8B (Dart-Prop)	MathVista-Overall ↑	both	≥4.4	38.10	38.20	+0.10	38.70	+0.60	36.30	-1.80	37.80	-0.30	38.30	+0.20	38.80	+0.70	37.30	-0.80
	MathVista-General (NEW) ↑	perception	≥6.5	38.95	39.09	+0.14	39.46	+0.51	36.98	-1.97	38.85	-0.10	39.16	+0.20	39.50	+0.54	37.93	-1.02
	MathVista-Math (NEW) ↑	reasoning	≥6.0	25.74	29.44	+3.70	31.30	+5.56	27.96	+2.22	28.52	+2.78	27.41	+1.67	29.07	+3.33	26.30	+0.56
	MathVerse-Overall ↑	reasoning	≥4.9	18.55	—	—	18.76	+0.20	20.10	+1.55	20.10	+1.55	19.62	+1.07	18.93	+0.38	18.83	+0.28
	MathVerse-T-D ↑	reasoning	≥4.9	22.59	26.78	+4.19	24.62	+2.03	26.02	+3.43	25.76	+3.17	25.25	+2.66	24.11	+1.52	22.59	+0.00
	MathVerse-T-L ↑	reasoning	≥4.9	19.80	18.53	-1.27	20.69	+0.89	21.70	+1.90	20.81	+1.01	21.32	+1.52	20.05	+0.25	20.56	+0.76
	MathVerse-V-I ↑	reasoning	≥4.9	19.80	—	—	18.91	-0.89	21.83	+2.03	22.59	+2.79	20.94	+1.14	19.80	+0.00	20.56	+0.76
	MathVerse-V-D ↑	reasoning	≥4.9	15.61	16.24	+0.63	16.88	+1.27	16.37	+0.76	16.24	+0.63	15.23	-0.38	15.86	+0.25	15.99	+0.38
	MathVerse-V-O ↑	reasoning	≥4.9	14.97	10.03	-4.94	12.69	-2.28	14.59	-0.38	15.10	+0.13	15.36	+0.39	14.85	-0.12	14.47	-0.50
	MathVision ↑	reasoning	≥7.9	14.80	13.82	-0.99	13.16	-1.64	13.49	-1.32	13.82	-0.99	15.13	+0.33	15.79	+0.99	16.78	+1.97
Idefics2-8B (MAmmoTH-1)	MMStar-Overall ↑	both	≥3.6	43.33	41.80	-1.53	42.87	-0.47	43.53	+0.20	43.73	+0.40	43.47	+0.13	44.00	+0.67	43.47	+0.13
	MMStar-Math ↑	reasoning	≥8.8	29.60	30.00	+0.40	29.20	-0.40	30.80	+1.20	32.80	+3.20	33.60	+4.00	32.80	+3.20	31.20	+1.60
	DynaMath-Average ↑	reasoning	≥2.0	22.61	24.37	+1.76	25.47	+2.85	22.24	-0.38	24.15	+1.54	24.01	+1.40	23.73	+1.12	22.52	-0.10
	DynaMath-Worst (NEW) ↑	reasoning	≥6.2	3.39	2.20	-1.20	2.59	-0.80	3.99	+0.60	4.19	+0.80	4.19	+0.80	3.79	+0.40	3.19	-0.20
	MathVista-Overall ↑	both	≥4.4	52.00	48.10	-3.90	49.50	-2.50	49.70	-2.30	50.50	-1.50	50.60	-1.40	51.50	-0.50	—	—
	MathVista-General (NEW) ↑	perception	≥6.5	51.04	47.33	-3.71	48.79	-2.25	49.00	-2.04	49.64	-1.40	49.74	-1.29	50.60	-0.44	—	—
	MathVista-Math (NEW) ↑	reasoning	≥6.0	48.33	42.41	-5.93	43.70	-4.63	44.26	-4.07	45.74	-2.59	45.93	-2.41	46.85	-1.48	—	—
	MathVerse-Overall ↑	reasoning	≥4.9	18.17	18.20	+0.02	18.68	+0.51	17.72	-0.46	17.87	-0.31	17.84	-0.33	18.63	+0.45	—	—
	MathVerse-T-D ↑	reasoning	≥4.9	21.83	22.08	+0.25	22.59	+0.76	21.83	+0.00	20.81	-1.02	20.94	-0.89	22.84	+1.01	—	—
	MathVerse-T-L ↑	reasoning	≥4.9	20.18	20.18	+0.00	21.07	+0.89	19.80	-0.38	19.92	-0.26	19.92	-0.26	20.81	+0.63	—	—
Qwen2-VL-7B-Instruct (Qwen2-Math-7B)	MathVerse-V-I ↑	reasoning	≥4.9	19.54	20.43	+0.89	20.69	+1.15	19.42	-0.12	19.42	-0.12	18.78	-0.76	19.67	+0.13	—	—
	MathVerse-V-D ↑	reasoning	≥4.9	19.04	18.02	-1.02	17.89	-1.15	17.13	-1.91	18.40	-0.64	18.91	-0.13	18.65	-0.39	—	—
	MathVerse-V-O ↑	reasoning	≥4.9	10.28	10.28	+0.00	11.17	+0.89	10.41	+0.13	10.79	+0.51	10.66	+0.38	11.17	+0.89	—	—
	MathVision ↑	reasoning	≥7.9	16.78	16.12	-0.66	18.75	+1.97	16.12	-0.66	13.82	-2.96	14.80	-1.97	14.47	-2.30	—	—
	MMStar-Overall ↑	both	≥3.6	49.33	46.73	-2.60	47.80	-1.53	47.67	-1.67	48.80	-0.53	48.47	-0.87	48.40	-0.93	—	—
	MMStar-Math ↑	reasoning	≥8.8	39.60	40.00	+0.40	40.40	+0.80	41.20	+1.60	41.60	+2.00	41.60	+2.00	39.60	+0.00	—	—
	DynaMath-Average ↑	reasoning	≥2.0	21.80	23.89	+2.10	22.85	+1.06	22.38	+0.58	22.24	+0.44	22.89	+1.10	22.53	+0.74	—	—
	DynaMath-Worst (NEW) ↑	reasoning	≥6.2	3.19	4.79	+1.60	4.19	+1.00	4.39	+1.20	3.59	+0.40	4.39	+1.20	3.79	+0.60	—	—
	MathVista-Overall ↑	both	≥4.4	61.10	23.70	-37.40	23.70	-37.40	30.70	-30.40	41.30	-19.80	52.80	-8.30	57.80	-3.30	—	—
	MathVista-General (NEW) ↑	perception	≥6.5	60.33	24.04	-36.30	24.00	-36.33	31.19	-29.15	40.89	-19.44	52.40	-7.93	57.03	-3.30	—	—
	MathVista-Math (NEW) ↑	reasoning	≥6.0	53.70	20.00	-33.70	19.26	-34.44	23.89	-29.81	33.15	-20.56	42.96	-10.74	50.00	-3.70	—	—
	MathVerse-Overall ↑	reasoning	≥4.9	30.68	0.00	-30.68	0.66	-30.02	6.40	-24.29	19.03	-11.65	25.00	-5.68	28.66	-2.03	—	—

				Base	$\alpha=0.3$		$\alpha=0.4$		$\alpha=0.5$		$\alpha=0.6$		$\alpha=0.7$		$\alpha=0.8$		$\alpha=0.9$	
Model	Benchmark	Type	Real Δ (pp)	Local	Local	Δ	Local	Δ	Local	Δ	Local	Δ	Local	Δ	Local	Δ	Local	Δ
	MMStar-Math $\uparrow$	reasoning	≥ 8.8	27.20	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	DynaMath-Average $\uparrow$	reasoning	≥ 2.0	17.78	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	DynaMath-Worst (NEW) $\uparrow$	reasoning	≥ 6.2	2.20	—	—	—	—	—	—	—	—	—	—	—	—	—	—
LLaVA-OneVision-7B (Qwen2-Math-7B)	MathVista-Overall $\uparrow$	both	≥ 4.4	—	25.90	—	31.00	—	42.20	—	51.20	—	58.90	—	61.10	—	62.50	—
	MathVista-General (NEW) $\uparrow$	perception	≥ 6.5	—	26.29	—	31.77	—	42.42	—	51.24	—	58.09	—	60.16	—	61.39	—
	MathVista-Math (NEW) $\uparrow$	reasoning	≥ 6.0	—	22.78	—	26.67	—	40.74	—	50.74	—	60.19	—	63.70	—	64.07	—
	MathVerse-Overall $\uparrow$	reasoning	≥ 4.9	—	—	—	10.46	—	21.17	—	27.18	—	30.15	—	30.97	—	30.74	—
	MathVerse-T-D $\uparrow$	reasoning	≥ 4.9	—	5.08	—	13.58	—	24.62	—	31.98	—	36.17	—	39.85	—	38.96	—
	MathVerse-T-L $\uparrow$	reasoning	≥ 4.9	—	—	—	12.94	—	23.35	—	29.44	—	33.25	—	34.77	—	33.38	—
	MathVerse-V-I $\uparrow$	reasoning	≥ 4.9	—	2.28	—	13.71	—	23.22	—	28.55	—	31.85	—	30.84	—	33.38	—
	MathVerse-V-D $\uparrow$	reasoning	≥ 4.9	—	2.92	—	10.28	—	20.94	—	28.17	—	30.58	—	30.84	—	29.44	—
	MathVerse-V-O $\uparrow$	reasoning	≥ 4.9	17.64	0.25	-17.39	1.78	-15.86	13.71	-3.93	17.77	+0.13	18.91	+1.27	18.53	+0.89	18.53	+0.89
	MathVision $\uparrow$	reasoning	≥ 7.9	18.42	10.20	-8.22	12.17	-6.25	12.50	-5.92	16.45	-1.97	17.43	-0.99	18.09	-0.33	18.42	+0.00
	MMStar-Overall $\uparrow$	both	≥ 3.6	61.93	3.53	-58.40	39.07	-22.87	49.60	-12.33	55.73	-6.20	59.40	-2.53	60.27	-1.67	61.07	-0.87
	MMStar-Math $\uparrow$	reasoning	≥ 8.8	58.80	1.20	-57.60	24.40	-34.40	36.00	-22.80	48.80	-10.00	57.20	-1.60	56.80	-2.00	57.20	-1.60
	DynaMath-Average $\uparrow$	reasoning	≥ 2.0	19.66	0.82	-18.84	10.56	-9.10	12.71	-6.95	12.71	-6.95	21.50	+1.84	23.21	+3.55	19.46	-0.20
	DynaMath-Worst (NEW) $\uparrow$	reasoning	≥ 6.2	4.99	0.00	-4.99	1.40	-3.59	2.59	-2.40	2.40	-2.59	4.59	-0.40	0.00	-4.99	4.59	-0.40
InternVL2.5-8B (InternLM2-Math-Plus-7B)	MathVista-Overall $\uparrow$	both	≥ 4.4	65.00	64.90	-0.10	64.60	-0.40	64.50	-0.50	64.50	-0.50	64.60	-0.40	64.40	-0.60	64.70	-0.30
	MathVista-General (NEW) $\uparrow$	perception	≥ 6.5	64.59	64.56	-0.03	64.22	-0.37	64.11	-0.48	64.11	-0.48	64.22	-0.37	64.01	-0.58	64.32	-0.27
	MathVista-Math (NEW) $\uparrow$	reasoning	≥ 6.0	61.48	60.19	-1.30	60.19	-1.30	60.37	-1.11	60.56	-0.93	60.37	-1.11	60.37	-1.11	60.19	-1.30
	MathVerse-Overall $\uparrow$	reasoning	≥ 4.9	34.67	34.80	+0.13	34.57	-0.10	34.90	+0.23	34.62	-0.05	34.90	+0.23	34.62	-0.05	34.90	+0.23
	MathVerse-T-D $\uparrow$	reasoning	≥ 4.9	42.89	43.65	+0.76	43.78	+0.89	44.29	+1.40	44.04	+1.15	43.91	+1.02	44.04	+1.15	43.53	+0.64
	MathVerse-T-L $\uparrow$	reasoning	≥ 4.9	38.58	37.44	-1.14	36.68	-1.90	36.80	-1.78	37.18	-1.40	37.18	-1.40	36.55	-2.03	37.94	-0.64
	MathVerse-V-I $\uparrow$	reasoning	≥ 4.9	37.31	37.82	+0.51	37.44	+0.13	37.44	+0.13	37.69	+0.38	37.18	-0.13	37.56	+0.25	37.82	+0.51
	MathVerse-V-D $\uparrow$	reasoning	≥ 4.9	33.25	33.38	+0.13	32.74	-0.51	33.76	+0.51	32.36	-0.89	34.14	+0.89	33.12	-0.13	33.50	+0.25
	MathVerse-V-O $\uparrow$	reasoning	≥ 4.9	21.32	21.70	+0.38	22.21	+0.89	22.21	+0.89	21.83	+0.51	22.08	+0.76	21.83	+0.51	21.70	+0.38
	MathVision $\uparrow$	reasoning	≥ 7.9	23.68	21.38	-2.30	16.45	-7.24	17.11	-6.58	18.42	-5.26	17.11	-6.58	21.71	-1.97	22.04	-1.64
	MMStar-Overall $\uparrow$	both	≥ 3.6	62.73	63.13	+0.40	63.00	+0.27	62.47	-0.27	62.47	-0.27	62.33	-0.40	62.47	-0.27	63.13	+0.40
	MMStar-Math $\uparrow$	reasoning	≥ 8.8	68.00	67.60	-0.40	69.20	+1.20	67.20	-0.80	67.20	-0.80	67.60	-0.40	67.20	-0.80	67.60	-0.40
	DynaMath-Average $\uparrow$	reasoning	≥ 2.0	32.22	32.16	-0.06	32.32	+0.10	32.32	+0.10	31.36	-0.86	32.30	+0.08	30.82	-1.40	32.16	-0.06
	DynaMath-Worst (NEW) $\uparrow$	reasoning	≥ 6.2	9.38	9.58	+0.20	9.38	+0.00	9.38	+0.00	8.38	-1.00	9.38	+0.00	8.78	-0.60	9.58	+0.20
Qwen2.5-VL-7B (Qwen2.5-Math-7B)	MathVista-Overall $\uparrow$	both	≥ 4.4	68.90	21.50	-47.40	29.00	-39.90	46.80	-22.10	55.50	-13.40	60.80	-8.10	64.20	-4.70	66.40	-2.50
	MathVista-General (NEW) $\uparrow$	perception	≥ 6.5	68.20	21.42	-46.78	29.42	-38.78	46.82	-21.38	55.57	-12.63	60.61	-7.59	64.04	-4.15	65.78	-2.42
	MathVista-Math (NEW) $\uparrow$	reasoning	≥ 6.0	67.04	19.81	-47.22	27.59	-39.44	41.11	-25.93	49.81	-17.22	55.56	-11.48	61.11	-5.93	64.44	-2.59

[illegible]

Masked alpha sweep — PMBT gate (pmbt_t*_v1.0_m1.0_mlponly_hgate)

Masked task-arithmetic on text-classified MLP neurons (v=m=1.0, mlp-only). Local = measured score; Δ = vs the model's Baseline, colored at each benchmark's Real Δ (pp) noise floor (yellow = favorable ≥ floor, red = unfavorable ≥ floor; orange/pink = below the floor). — = no score on disk. MathVerse-Overall = mean of 5 splits. Real Δ (pp) = min change beating the 95% noise floor (see final-table references); treat any Δ below it as noise.

				Base	α=0.3		α=0.4		α=0.5		α=0.6		α=0.7		α=0.8		α=0.9	
Model	Benchmark	Type	Real Δ (pp)	Local	Local	Δ	Local	Δ	Local	Δ	Local	Δ	Local	Δ	Local	Δ	Local	Δ
LLaVA-Next-LLaMA3-8B (Dart-Prop)	MathVista-Overall ↑	both	≥4.4	38.10	37.40	-0.70	38.10	+0.00	36.70	-1.40	37.00	-1.10	37.30	-0.80	39.20	+1.10	38.00	-0.10
	MathVista-General (NEW) ↑	perception	≥6.5	38.95	38.61	-0.34	39.09	+0.14	37.76	-1.19	38.13	-0.82	38.17	-0.78	40.07	+1.12	38.58	-0.37
	MathVista-Math (NEW) ↑	reasoning	≥6.0	25.74	29.63	+3.89	29.81	+4.07	27.96	+2.22	28.15	+2.41	27.59	+1.85	29.44	+3.70	27.96	+2.22
	MathVerse-Overall ↑	reasoning	≥4.9	18.55	18.12	-0.43	18.30	-0.26	19.47	+0.91	19.39	+0.84	19.34	+0.79	18.35	-0.21	18.91	+0.35
	MathVerse-T-D ↑	reasoning	≥4.9	22.59	26.27	+3.68	25.25	+2.66	26.02	+3.43	25.25	+2.66	25.76	+3.17	23.60	+1.01	22.97	+0.38
	MathVerse-T-L ↑	reasoning	≥4.9	19.80	19.04	-0.76	18.53	-1.27	21.57	+1.77	21.45	+1.65	20.94	+1.14	19.16	-0.64	20.30	+0.50
	MathVerse-V-I ↑	reasoning	≥4.9	19.80	17.39	-2.41	17.89	-1.91	20.56	+0.76	21.32	+1.52	21.07	+1.27	19.16	-0.64	20.81	+1.01
	MathVerse-V-D ↑	reasoning	≥4.9	15.61	16.75	+1.14	15.99	+0.38	15.61	+0.00	15.36	-0.25	14.34	-1.27	14.85	-0.76	15.99	+0.38
	MathVerse-V-O ↑	reasoning	≥4.9	14.97	11.17	-3.80	13.83	-1.14	13.58	-1.39	13.58	-1.39	14.59	-0.38	14.97	+0.00	14.47	-0.50
	MathVision ↑	reasoning	≥7.9	14.80	13.82	-0.99	10.86	-3.95	15.46	+0.66	16.12	+1.32	13.82	-0.99	13.49	-1.32	15.13	+0.33
Idefics2-8B (MAmmoTH-1)	MMStar-Overall ↑	both	≥3.6	43.33	38.87	-4.47	40.20	-3.13	41.87	-1.47	43.20	-0.13	43.53	+0.20	43.60	+0.27	43.80	+0.47
	MMStar-Math ↑	reasoning	≥8.8	29.60	33.20	+3.60	30.40	+0.80	31.20	+1.60	31.20	+1.60	32.80	+3.20	33.60	+4.00	32.40	+2.80
	DynaMath-Average ↑	reasoning	≥2.0	22.61	25.19	+2.57	23.53	+0.92	19.50	-3.11	23.07	+0.46	23.49	+0.88	22.42	-0.20	—	—
	DynaMath-Worst (NEW) ↑	reasoning	≥6.2	3.39	5.19	+1.80	2.20	-1.20	3.39	+0.00	3.99	+0.60	4.99	+1.60	4.19	+0.80	—	—
	MathVista-Overall ↑	both	≥4.4	52.00	49.50	-2.50	50.20	-1.80	50.80	-1.20	51.00	-1.00	50.70	-1.30	51.20	-0.80	—	—
	MathVista-General (NEW) ↑	perception	≥6.5	51.04	48.31	-2.72	49.23	-1.80	49.78	-1.26	49.95	-1.09	49.68	-1.36	50.22	-0.82	—	—
	MathVista-Math (NEW) ↑	reasoning	≥6.0	48.33	46.48	-1.85	45.74	-2.59	45.93	-2.41	46.67	-1.67	45.74	-2.59	46.30	-2.04	—	—
	MathVerse-Overall ↑	reasoning	≥4.9	18.17	18.35	+0.18	17.67	-0.51	17.95	-0.23	18.17	+0.00	17.97	-0.21	18.35	+0.18	—	—
	MathVerse-T-D ↑	reasoning	≥4.9	21.83	22.08	+0.25	21.57	-0.26	22.21	+0.38	22.46	+0.63	21.95	+0.12	22.59	+0.76	—	—
	MathVerse-T-L ↑	reasoning	≥4.9	20.18	20.81	+0.63	19.80	-0.38	19.42	-0.76	19.92	-0.26	19.92	-0.26	20.43	+0.25	—	—
Qwen2-VL-7B-Instruct (Qwen2-Math-7B)	MathVerse-V-I ↑	reasoning	≥4.9	19.54	20.94	+1.40	19.29	-0.25	19.42	-0.12	19.04	-0.50	18.65	-0.89	18.78	-0.76	—	—
	MathVerse-V-D ↑	reasoning	≥4.9	19.04	18.02	-1.02	16.88	-2.16	17.13	-1.91	17.77	-1.27	18.40	-0.64	18.91	-0.13	—	—
	MathVerse-V-O ↑	reasoning	≥4.9	10.28	9.90	-0.38	10.79	+0.51	11.55	+1.27	11.68	+1.40	10.91	+0.63	11.04	+0.76	—	—
	MathVision ↑	reasoning	≥7.9	16.78	17.11	+0.33	16.12	-0.66	16.12	-0.66	16.45	-0.33	15.13	-1.64	14.47	-2.30	—	—
	MMStar-Overall ↑	both	≥3.6	49.33	46.00	-3.33	46.60	-2.73	47.60	-1.73	48.20	-1.13	48.60	-0.73	49.07	-0.27	—	—
	MMStar-Math ↑	reasoning	≥8.8	39.60	40.40	+0.80	39.60	+0.00	40.00	+0.40	39.60	+0.00	39.60	+0.00	40.00	+0.40	—	—
	DynaMath-Average ↑	reasoning	≥2.0	21.80	21.72	-0.08	22.08	+0.28	21.32	-0.48	21.10	-0.70	21.22	-0.58	—	—	—	—
	DynaMath-Worst (NEW) ↑	reasoning	≥6.2	3.19	4.39	+1.20	3.99	+0.80	4.79	+1.60	3.79	+0.60	4.19	+1.00	—	—	—	—
	MathVista-Overall ↑	both	≥4.4	61.10	24.30	-36.80	23.70	-37.40	30.90	-30.20	44.80	-16.30	51.10	-10.00	57.90	-3.20	—	—
	MathVista-General (NEW) ↑	perception	≥6.5	60.33	24.55	-35.78	24.14	-36.19	31.15	-29.18	44.60	-15.73	50.63	-9.70	57.30	-3.03	—	—
	MathVista-Math (NEW) ↑	reasoning	≥6.0	53.70	21.85	-31.85	20.74	-32.96	24.07	-29.63	35.93	-17.78	42.78	-10.93	49.63	-4.07	—	—
	MathVerse-Overall ↑	reasoning	≥4.9	30.68	0.00	-30.68	0.48	-30.20	3.12	-27.56	17.16	-13.53	24.80	-5.89	28.63	-2.05	—	—

				Base	$\alpha=0.3$		$\alpha=0.4$		$\alpha=0.5$		$\alpha=0.6$		$\alpha=0.7$		$\alpha=0.8$		$\alpha=0.9$	
Model	Benchmark	Type	Real Δ (pp)	Local	Local	Δ	Local	Δ	Local	Δ	Local	Δ	Local	Δ	Local	Δ	Local	Δ
	MMStar-Math $\uparrow$	reasoning	≥ 8.8	27.20	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	DynaMath-Average $\uparrow$	reasoning	≥ 2.0	17.78	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	DynaMath-Worst (NEW) $\uparrow$	reasoning	≥ 6.2	2.20	—	—	—	—	—	—	—	—	—	—	—	—	—	—
LLaVA-OneVision-7B (Qwen2-Math-7B)	MathVista-Overall $\uparrow$	both	≥ 4.4	—	25.60	—	28.60	—	36.00	—	49.20	—	60.30	—	61.60	—	62.80	—
	MathVista-General (NEW) $\uparrow$	perception	≥ 6.5	—	25.88	—	29.21	—	36.33	—	49.23	—	59.72	—	60.78	—	61.87	—
	MathVista-Math (NEW) $\uparrow$	reasoning	≥ 6.0	—	22.41	—	22.78	—	31.48	—	46.85	—	60.74	—	62.59	—	64.26	—
	MathVerse-Overall $\uparrow$	reasoning	≥ 4.9	—	5.87	—	9.92	—	18.50	—	26.55	—	29.39	—	30.76	—	30.48	—
	MathVerse-T-D $\uparrow$	reasoning	≥ 4.9	—	12.06	—	15.36	—	21.45	—	31.98	—	37.06	—	39.09	—	39.97	—
	MathVerse-T-L $\uparrow$	reasoning	≥ 4.9	—	6.22	—	11.80	—	20.81	—	28.43	—	32.61	—	34.64	—	34.01	—
	MathVerse-V-I $\uparrow$	reasoning	≥ 4.9	—	5.46	—	12.69	—	20.30	—	27.28	—	30.84	—	31.98	—	32.23	—
	MathVerse-V-D $\uparrow$	reasoning	≥ 4.9	—	3.81	—	9.39	—	19.80	—	26.78	—	28.81	—	30.20	—	28.30	—
	MathVerse-V-O $\uparrow$	reasoning	≥ 4.9	17.64	1.78	-15.86	0.38	-17.26	10.15	-7.49	18.27	+0.63	17.64	+0.00	17.89	+0.25	17.89	+0.25
	MathVision $\uparrow$	reasoning	≥ 7.9	18.42	7.24	-11.18	10.20	-8.22	13.16	-5.26	17.76	-0.66	17.43	-0.99	17.43	-0.99	16.45	-1.97
	MMStar-Overall $\uparrow$	both	≥ 3.6	61.93	3.40	-58.53	15.73	-46.20	44.53	-17.40	53.53	-8.40	58.00	-3.93	60.73	-1.20	61.27	-0.67
	MMStar-Math $\uparrow$	reasoning	≥ 8.8	58.80	3.20	-55.60	7.60	-51.20	32.80	-26.00	43.20	-15.60	52.80	-6.00	56.00	-2.80	57.20	-1.60
	DynaMath-Average $\uparrow$	reasoning	≥ 2.0	19.66	0.82	-18.84	2.71	-16.95	6.81	-12.85	21.74	+2.08	23.75	+4.09	27.92	+8.26	19.30	-0.36
	DynaMath-Worst (NEW) $\uparrow$	reasoning	≥ 6.2	4.99	0.00	-4.99	0.00	-4.99	1.40	-3.59	4.39	-0.60	1.00	-3.99	4.79	-0.20	4.39	-0.60
InternVL2.5-8B (InternLM2-Math-Plus-7B)	MathVista-Overall $\uparrow$	both	≥ 4.4	65.00	64.60	-0.40	64.90	-0.10	64.70	-0.30	64.60	-0.40	64.80	-0.20	64.60	-0.40	64.70	-0.30
	MathVista-General (NEW) $\uparrow$	perception	≥ 6.5	64.59	64.25	-0.34	64.56	-0.03	64.32	-0.27	64.22	-0.37	64.42	-0.17	64.22	-0.37	64.32	-0.27
	MathVista-Math (NEW) $\uparrow$	reasoning	≥ 6.0	61.48	60.00	-1.48	60.19	-1.30	60.56	-0.93	60.56	-0.93	60.19	-1.30	60.37	-1.11	60.19	-1.30
	MathVerse-Overall $\uparrow$	reasoning	≥ 4.9	34.67	34.54	-0.13	34.75	+0.08	34.72	+0.05	34.72	+0.05	35.38	+0.71	34.49	-0.18	34.75	+0.08
	MathVerse-T-D $\uparrow$	reasoning	≥ 4.9	42.89	43.91	+1.02	44.29	+1.40	44.29	+1.40	44.67	+1.78	45.05	+2.16	43.91	+1.02	43.53	+0.64
	MathVerse-T-L $\uparrow$	reasoning	≥ 4.9	38.58	36.55	-2.03	37.56	-1.02	37.44	-1.14	37.06	-1.52	38.20	-0.38	36.93	-1.65	37.82	-0.76
	MathVerse-V-I $\uparrow$	reasoning	≥ 4.9	37.31	37.18	-0.13	37.31	+0.00	37.31	+0.00	37.18	-0.13	37.44	+0.13	37.06	-0.25	37.31	+0.00
	MathVerse-V-D $\uparrow$	reasoning	≥ 4.9	33.25	33.25	+0.00	32.87	-0.38	32.74	-0.51	32.49	-0.76	34.14	+0.89	32.49	-0.76	33.38	+0.13
	MathVerse-V-O $\uparrow$	reasoning	≥ 4.9	21.32	21.83	+0.51	21.70	+0.38	21.83	+0.51	22.21	+0.89	22.08	+0.76	22.08	+0.76	21.70	+0.38
	MathVision $\uparrow$	reasoning	≥ 7.9	23.68	21.71	-1.97	18.09	-5.59	18.09	-5.59	18.09	-5.59	17.43	-6.25	18.09	-5.59	21.71	-1.97
	MMStar-Overall $\uparrow$	both	≥ 3.6	62.73	62.47	-0.27	63.13	+0.40	63.13	+0.40	62.33	-0.40	62.47	-0.27	62.47	-0.27	63.13	+0.40
	MMStar-Math $\uparrow$	reasoning	≥ 8.8	68.00	67.20	-0.80	67.60	-0.40	67.60	-0.40	67.60	-0.40	67.20	-0.80	67.20	-0.80	67.60	-0.40
	DynaMath-Average $\uparrow$	reasoning	≥ 2.0	32.22	32.18	-0.04	32.30	+0.08	32.30	+0.08	32.26	+0.04	30.88	-1.34	32.26	+0.04	32.16	-0.06
	DynaMath-Worst (NEW) $\uparrow$	reasoning	≥ 6.2	9.38	9.58	+0.20	9.38	+0.00	9.38	+0.00	9.58	+0.20	8.58	-0.80	9.58	+0.20	9.58	+0.20
Qwen2.5-VL-7B (Qwen2.5-Math-7B)	MathVista-Overall $\uparrow$	both	≥ 4.4	68.90	23.60	-45.30	25.30	-43.60	42.10	-26.80	53.30	-15.60	59.60	-9.30	64.50	-4.40	65.60	-3.30
	MathVista-General (NEW) $\uparrow$	perception	≥ 6.5	68.20	23.80	-44.40	25.20	-43.00	41.81	-26.39	53.35	-14.85	59.28	-8.92	64.11	-4.09	65.13	-3.06
	MathVista-Math (NEW) $\uparrow$	reasoning	≥ 6.0	67.04	20.93	-46.11	24.07	-42.96	38.15	-28.89	46.30	-20.74	55.37	-11.67	60.93	-6.11	62.78	-4.26

[illegible]

Masked alpha sweep — FT gate (ft_pmbt_t*_v1.0_m1.0_mlponly_hgate)

Masked task-arithmetic on text-classified MLP neurons (v=m=1.0, mlp-only). Local = measured score; Δ = vs the model's Baseline, colored at each benchmark's Real Δ (pp) noise floor (yellow = favorable ≥ floor, red = unfavorable ≥ floor; orange/pink = below the floor). — = no score on disk. MathVerse-Overall = mean of 5 splits. Real Δ (pp) = min change beating the 95% noise floor (see final-table references); treat any Δ below it as noise.

				Base	α=0.3		α=0.4		α=0.5		α=0.6		α=0.7		α=0.8		α=0.9	
Model	Benchmark	Type	Real Δ (pp)	Local	Local	Δ	Local	Δ	Local	Δ	Local	Δ	Local	Δ	Local	Δ	Local	Δ
LLaVA-Next-LLaMA3-8B (Dart-Prop)	MathVista-Overall ↑	both	≥4.4	38.10	36.20	-1.90	38.00	-0.10	39.40	+1.30	39.30	+1.20	38.10	+0.00	37.60	-0.50	37.60	-0.50
	MathVista-General (NEW) ↑	perception	≥6.5	38.95	36.98	-1.97	38.85	-0.10	40.25	+1.29	40.04	+1.09	38.68	-0.27	38.17	-0.78	38.37	-0.58
	MathVista-Math (NEW) ↑	reasoning	≥6.0	25.74	27.96	+2.22	28.15	+2.41	29.07	+3.33	29.81	+4.07	28.52	+2.78	27.41	+1.67	25.00	-0.74
	MathVerse-Overall ↑	reasoning	≥4.9	18.55	19.04	+0.48	—	—	18.93	+0.38	18.45	-0.10	19.19	+0.63	18.68	+0.12	18.73	+0.18
	MathVerse-T-D ↑	reasoning	≥4.9	22.59	24.37	+1.78	24.49	+1.90	23.48	+0.89	23.48	+0.89	23.22	+0.63	22.21	-0.38	22.34	-0.25
	MathVerse-T-L ↑	reasoning	≥4.9	19.80	20.05	+0.25	19.16	-0.64	20.56	+0.76	19.80	+0.00	20.43	+0.63	20.30	+0.50	20.30	+0.50
	MathVerse-V-I ↑	reasoning	≥4.9	19.80	19.80	+0.00	19.54	-0.26	19.92	+0.12	18.78	-1.02	21.57	+1.77	20.43	+0.63	20.30	+0.50
	MathVerse-V-D ↑	reasoning	≥4.9	15.61	15.23	-0.38	—	—	15.23	-0.38	15.10	-0.51	16.12	+0.51	15.86	+0.25	15.74	+0.13
	MathVerse-V-O ↑	reasoning	≥4.9	14.97	15.74	+0.77	16.12	+1.15	15.48	+0.51	15.10	+0.13	14.59	-0.38	14.59	-0.38	14.97	+0.00
	MathVision ↑	reasoning	≥7.9	14.80	14.14	-0.66	13.82	-0.99	13.49	-1.32	12.17	-2.63	13.16	-1.64	13.82	-0.99	14.14	-0.66
Idefics2-8B (MAmmoTH-1)	MMStar-Overall ↑	both	≥3.6	43.33	42.27	-1.07	43.27	-0.07	43.33	+0.00	43.40	+0.07	43.33	+0.00	43.13	-0.20	43.60	+0.27
	MMStar-Math ↑	reasoning	≥8.8	29.60	30.00	+0.40	30.80	+1.20	32.00	+2.40	32.40	+2.80	32.80	+3.20	31.20	+1.60	32.00	+2.40
	DynaMath-Average ↑	reasoning	≥2.0	22.61	22.34	-0.28	21.92	-0.70	21.54	-1.08	20.84	-1.78	21.06	-1.56	21.48	-1.14	—	—
	DynaMath-Worst (NEW) ↑	reasoning	≥6.2	3.39	4.19	+0.80	3.79	+0.40	3.39	+0.00	3.19	-0.20	2.59	-0.80	2.79	-0.60	—	—
	MathVista-Overall ↑	both	≥4.4	52.00	50.90	-1.10	51.20	-0.80	52.40	+0.40	51.60	-0.40	51.60	-0.40	51.40	-0.60	—	—
	MathVista-General (NEW) ↑	perception	≥6.5	51.04	50.05	-0.99	50.32	-0.71	51.58	+0.54	50.70	-0.34	50.70	-0.34	50.49	-0.54	—	—
	MathVista-Math (NEW) ↑	reasoning	≥6.0	48.33	46.85	-1.48	47.41	-0.93	48.15	-0.19	47.59	-0.74	47.59	-0.74	47.41	-0.93	—	—
	MathVerse-Overall ↑	reasoning	≥4.9	18.17	17.49	-0.69	18.02	-0.15	17.59	-0.59	17.59	-0.59	17.84	-0.33	18.05	-0.13	—	—
	MathVerse-T-D ↑	reasoning	≥4.9	21.83	21.32	-0.51	22.21	+0.38	21.83	+0.00	21.95	+0.12	22.59	+0.76	22.59	+0.76	—	—
	MathVerse-T-L ↑	reasoning	≥4.9	20.18	20.18	+0.00	20.43	+0.25	19.67	-0.51	19.29	-0.89	19.80	-0.38	19.42	-0.76	—	—
Qwen2-VL-7B-Instruct (Qwen2-Math-7B)	MathVerse-V-I ↑	reasoning	≥4.9	19.54	18.53	-1.01	19.42	-0.12	18.27	-1.27	18.40	-1.14	18.78	-0.76	19.29	-0.25	—	—
	MathVerse-V-D ↑	reasoning	≥4.9	19.04	18.02	-1.02	18.15	-0.89	17.51	-1.53	17.51	-1.53	17.51	-1.53	17.89	-1.15	—	—
	MathVerse-V-O ↑	reasoning	≥4.9	10.28	9.39	-0.89	9.90	-0.38	10.66	+0.38	10.79	+0.51	10.53	+0.25	11.04	+0.76	—	—
	MathVision ↑	reasoning	≥7.9	16.78	16.12	-0.66	15.79	-0.99	16.45	-0.33	16.78	+0.00	13.82	-2.96	15.13	-1.64	—	—
	MMStar-Overall ↑	both	≥3.6	49.33	47.80	-1.53	48.27	-1.07	48.40	-0.93	49.20	-0.13	49.27	-0.07	49.60	+0.27	—	—
	MMStar-Math ↑	reasoning	≥8.8	39.60	41.20	+1.60	40.80	+1.20	41.20	+1.60	40.80	+1.20	40.80	+1.20	41.60	+2.00	—	—
	DynaMath-Average ↑	reasoning	≥2.0	21.80	20.96	-0.84	20.58	-1.22	20.06	-1.74	19.78	-2.02	19.36	-2.44	18.52	-3.27	—	—
	DynaMath-Worst (NEW) ↑	reasoning	≥6.2	3.19	3.79	+0.60	4.19	+1.00	4.39	+1.20	4.39	+1.20	3.79	+0.60	3.39	+0.20	—	—
	MathVista-Overall ↑	both	≥4.4	61.10	28.50	-32.60	36.10	-25.00	45.40	-15.70	53.70	-7.40	57.70	-3.40	59.30	-1.80	—	—
	MathVista-General (NEW) ↑	perception	≥6.5	60.33	29.08	-31.26	36.06	-24.28	45.39	-14.95	53.56	-6.78	57.13	-3.20	58.63	-1.70	—	—
	MathVista-Math (NEW) ↑	reasoning	≥6.0	53.70	22.59	-31.11	30.19	-23.52	34.44	-19.26	44.07	-9.63	49.07	-4.63	51.67	-2.04	—	—
	MathVerse-Overall ↑	reasoning	≥4.9	30.68	2.18	-28.50	4.80	-25.89	15.68	-15.00	25.13	-5.56	27.99	-2.69	30.13	-0.56	—	—

				Base	$\alpha=0.3$		$\alpha=0.4$		$\alpha=0.5$		$\alpha=0.6$		$\alpha=0.7$		$\alpha=0.8$		$\alpha=0.9$	
Model	Benchmark	Type	Real Δ (pp)	Local	Local	Δ	Local	Δ	Local	Δ	Local	Δ	Local	Δ	Local	Δ	Local	Δ
	MMStar-Math $\uparrow$	reasoning	≥ 8.8	27.20	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	DynaMath-Average $\uparrow$	reasoning	≥ 2.0	17.78	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	DynaMath-Worst (NEW) $\uparrow$	reasoning	≥ 6.2	2.20	—	—	—	—	—	—	—	—	—	—	—	—	—	—
LLaVA-OneVision-7B (Qwen2-Math-7B)	MathVista-Overall $\uparrow$	both	≥ 4.4	—	52.80	—	58.00	—	61.70	—	62.50	—	62.50	—	62.90	—	62.90	—
	MathVista-General (NEW) $\uparrow$	perception	≥ 6.5	—	52.23	—	57.61	—	61.08	—	61.87	—	61.70	—	61.97	—	61.90	—
	MathVista-Math (NEW) $\uparrow$	reasoning	≥ 6.0	—	51.85	—	55.93	—	60.19	—	61.30	—	62.22	—	64.07	—	63.70	—
	MathVerse-Overall $\uparrow$	reasoning	≥ 4.9	—	25.03	—	27.92	—	29.44	—	29.77	—	30.41	—	30.53	—	30.61	—
	MathVerse-T-D $\uparrow$	reasoning	≥ 4.9	—	30.71	—	34.39	—	35.03	—	37.06	—	37.31	—	38.71	—	39.47	—
	MathVerse-T-L $\uparrow$	reasoning	≥ 4.9	—	29.57	—	30.96	—	32.87	—	32.99	—	34.01	—	33.63	—	33.63	—
	MathVerse-V-I $\uparrow$	reasoning	≥ 4.9	—	28.68	—	29.82	—	31.98	—	31.47	—	32.23	—	32.23	—	32.11	—
	MathVerse-V-D $\uparrow$	reasoning	≥ 4.9	—	26.40	—	27.92	—	29.70	—	28.93	—	30.08	—	29.70	—	29.06	—
	MathVerse-V-O $\uparrow$	reasoning	≥ 4.9	17.64	9.77	-7.87	16.50	-1.14	17.64	+0.00	18.40	+0.76	18.40	+0.76	18.40	+0.76	18.78	+1.14
	MathVision $\uparrow$	reasoning	≥ 7.9	18.42	17.11	-1.32	17.43	-0.99	19.08	+0.66	17.43	-0.99	18.09	-0.33	18.42	+0.00	17.76	-0.66
	MMStar-Overall $\uparrow$	both	≥ 3.6	61.93	56.80	-5.13	59.07	-2.87	59.87	-2.07	60.80	-1.13	61.00	-0.93	60.87	-1.07	61.33	-0.60
	MMStar-Math $\uparrow$	reasoning	≥ 8.8	58.80	47.60	-11.20	54.00	-4.80	56.00	-2.80	58.40	-0.40	58.00	-0.80	58.00	-0.80	58.80	+0.00
	DynaMath-Average $\uparrow$	reasoning	≥ 2.0	19.66	4.43	-15.23	19.32	-0.34	26.31	+6.65	29.94	+10.28	30.80	+11.14	32.30	+12.63	19.44	-0.22
	DynaMath-Worst (NEW) $\uparrow$	reasoning	≥ 6.2	4.99	0.20	-4.79	2.59	-2.40	5.39	+0.40	7.58	+2.59	7.58	+2.59	7.98	+2.99	5.59	+0.60
InternVL2.5-8B (InternLM2-Math-Plus-7B)	MathVista-Overall $\uparrow$	both	≥ 4.4	65.00	66.00	+1.00	64.40	-0.60	64.70	-0.30	64.50	-0.50	64.50	-0.50	65.90	+0.90	64.60	-0.40
	MathVista-General (NEW) $\uparrow$	perception	≥ 6.5	64.59	65.44	+0.85	64.01	-0.58	64.32	-0.27	64.11	-0.48	64.11	-0.48	65.34	+0.75	64.22	-0.37
	MathVista-Math (NEW) $\uparrow$	reasoning	≥ 6.0	61.48	63.70	+2.22	60.00	-1.48	60.56	-0.93	60.00	-1.48	60.56	-0.93	63.33	+1.85	60.00	-1.48
	MathVerse-Overall $\uparrow$	reasoning	≥ 4.9	34.67	34.85	+0.18	34.87	+0.20	34.95	+0.28	34.92	+0.25	34.65	-0.02	34.80	+0.13	34.75	+0.08
	MathVerse-T-D $\uparrow$	reasoning	≥ 4.9	42.89	44.29	+1.40	43.91	+1.02	44.42	+1.53	43.53	+0.64	44.42	+1.53	44.04	+1.15	43.78	+0.89
	MathVerse-T-L $\uparrow$	reasoning	≥ 4.9	38.58	37.94	-0.64	37.82	-0.76	37.18	-1.40	37.69	-0.89	37.44	-1.14	36.55	-2.03	37.94	-0.64
	MathVerse-V-I $\uparrow$	reasoning	≥ 4.9	37.31	37.31	+0.00	37.18	-0.13	37.56	+0.25	37.18	-0.13	37.18	-0.13	37.69	+0.38	37.06	-0.25
	MathVerse-V-D $\uparrow$	reasoning	≥ 4.9	33.25	33.12	-0.13	33.38	+0.13	33.50	+0.25	34.39	+1.14	32.49	-0.76	33.50	+0.25	33.12	-0.13
	MathVerse-V-O $\uparrow$	reasoning	≥ 4.9	21.32	21.57	+0.25	22.08	+0.76	22.08	+0.76	21.83	+0.51	21.70	+0.38	22.21	+0.89	21.83	+0.51
	MathVision $\uparrow$	reasoning	≥ 7.9	23.68	22.04	-1.64	22.04	-1.64	18.09	-5.59	18.09	-5.59	17.11	-6.58	17.43	-6.25	22.37	-1.32
	MMStar-Overall $\uparrow$	both	≥ 3.6	62.73	63.13	+0.40	63.13	+0.40	62.47	-0.27	63.13	+0.40	62.33	-0.40	62.47	-0.27	63.13	+0.40
	MMStar-Math $\uparrow$	reasoning	≥ 8.8	68.00	67.60	-0.40	67.60	-0.40	67.20	-0.80	67.60	-0.40	67.60	-0.40	67.20	-0.80	67.60	-0.40
	DynaMath-Average $\uparrow$	reasoning	≥ 2.0	32.22	32.30	+0.08	32.26	+0.04	32.30	+0.08	30.88	-1.34	30.82	-1.40	31.76	-0.46	32.16	-0.06
	DynaMath-Worst (NEW) $\uparrow$	reasoning	≥ 6.2	9.38	9.38	+0.00	9.58	+0.20	9.38	+0.00	8.58	-0.80	8.38	-1.00	8.58	-0.80	9.58	+0.20
Qwen2.5-VL-7B (Qwen2.5-Math-7B)	MathVista-Overall $\uparrow$	both	≥ 4.4	68.90	38.00	-30.90	51.90	-17.00	58.50	-10.40	61.60	-7.30	62.20	-6.70	64.20	-4.70	66.80	-2.10
	MathVista-General (NEW) $\uparrow$	perception	≥ 6.5	68.20	38.13	-30.06	51.86	-16.34	58.22	-9.98	61.18	-7.01	61.83	-6.37	64.15	-4.05	66.43	-1.77
	MathVista-Math (NEW) $\uparrow$	reasoning	≥ 6.0	67.04	32.22	-34.81	44.07	-22.96	51.67	-15.37	57.04	-10.00	57.78	-9.26	60.37	-6.67	63.52	-3.52

				Base	$\alpha=0.3$		$\alpha=0.4$		$\alpha=0.5$		$\alpha=0.6$		$\alpha=0.7$		$\alpha=0.8$		$\alpha=0.9$	
Model	Benchmark	Type	Real Δ (pp)	Local	Local	Δ	Local	Δ	Local	Δ	Local	Δ	Local	Δ	Local	Δ	Local	Δ
	MathVerse-Overall $\uparrow$	reasoning	≥ 4.9	44.01	8.27	-35.74	15.79	-28.22	24.95	-19.06	33.76	-10.26	39.01	-5.00	41.68	-2.34	43.20	-0.82
	MathVerse-T-D $\uparrow$	reasoning	≥ 4.9	53.43	6.09	-47.34	19.42	-34.01	31.47	-21.96	41.24	-12.19	48.86	-4.57	50.76	-2.67	51.52	-1.91
	MathVerse-T-L $\uparrow$	reasoning	≥ 4.9	45.18	7.61	-37.57	15.86	-29.32	25.13	-20.05	34.77	-10.41	38.20	-6.98	41.75	-3.43	43.91	-1.27
	MathVerse-V-I $\uparrow$	reasoning	≥ 4.9	40.61	8.12	-32.49	13.71	-26.90	22.08	-18.53	31.35	-9.26	36.93	-3.68	40.36	-0.25	40.86	+0.25
	MathVerse-V-D $\uparrow$	reasoning	≥ 4.9	40.10	7.49	-32.61	15.74	-24.36	23.60	-16.50	31.98	-8.12	36.04	-4.06	38.71	-1.39	40.48	+0.38
	MathVerse-V-O $\uparrow$	reasoning	≥ 4.9	40.74	12.06	-28.68	14.21	-26.53	22.46	-18.28	29.44	-11.30	35.03	-5.71	36.80	-3.94	39.21	-1.53
	MathVision $\uparrow$	reasoning	≥ 7.9	26.32	11.84	-14.47	15.46	-10.86	20.39	-5.92	24.01	-2.30	22.70	-3.62	25.66	-0.66	25.00	-1.32
	MMStar-Overall $\uparrow$	both	≥ 3.6	61.67	35.20	-26.47	48.53	-13.13	55.00	-6.67	58.60	-3.07	59.60	-2.07	61.33	-0.33	61.47	-0.20
	MMStar-Math $\uparrow$	reasoning	≥ 8.8	61.20	20.80	-40.40	43.60	-17.60	51.20	-10.00	55.20	-6.00	58.40	-2.80	60.40	-0.80	62.80	+1.60
	DynaMath-Average $\uparrow$	reasoning	≥ 2.0	53.27	17.90	-35.37	28.78	-24.49	39.58	-13.69	43.65	-9.62	47.07	-6.21	51.78	-1.50	53.67	+0.40
	DynaMath-Worst (NEW) $\uparrow$	reasoning	≥ 6.2	22.36	1.40	-20.96	4.59	-17.76	9.38	-12.97	12.77	-9.58	15.57	-6.79	21.36	-1.00	20.96	-1.40
Qwen2.5-VL-3B (Qwen2.5-3B-DAPO-math-reasoning)	MathVista-Overall $\uparrow$	both	≥ 4.4	61.80	—	—	—	—	—	—	—	—	—	—	—	—	61.40	-0.40
	MathVista-General (NEW) $\uparrow$	perception	≥ 6.5	61.46	—	—	—	—	—	—	—	—	—	—	—	—	61.05	-0.41
	MathVista-Math (NEW) $\uparrow$	reasoning	≥ 6.0	57.78	—	—	—	—	—	—	—	—	—	—	—	—	56.11	-1.67
	MathVerse-Overall $\uparrow$	reasoning	≥ 4.9	34.32	—	—	—	—	—	—	—	—	—	—	—	—	32.92	-1.40
	MathVerse-T-D $\uparrow$	reasoning	≥ 4.9	41.75	—	—	—	—	—	—	—	—	—	—	—	—	39.47	-2.28
	MathVerse-T-L $\uparrow$	reasoning	≥ 4.9	35.03	—	—	—	—	—	—	—	—	—	—	—	—	32.61	-2.42
	MathVerse-V-I $\uparrow$	reasoning	≥ 4.9	32.11	—	—	—	—	—	—	—	—	—	—	—	—	31.73	-0.38
	MathVerse-V-D $\uparrow$	reasoning	≥ 4.9	33.12	—	—	—	—	—	—	—	—	—	—	—	—	31.85	-1.27
	MathVerse-V-O $\uparrow$	reasoning	≥ 4.9	29.57	—	—	—	—	—	—	—	—	—	—	—	—	28.93	-0.64
	MathVision $\uparrow$	reasoning	≥ 7.9	23.36	—	—	—	—	—	—	—	—	—	—	—	—	23.68	+0.33
	MMStar-Overall $\uparrow$	both	≥ 3.6	53.93	—	—	—	—	—	—	—	—	—	—	—	—	54.60	+0.67
	MMStar-Math $\uparrow$	reasoning	≥ 8.8	58.00	—	—	—	—	—	—	—	—	—	—	—	—	58.00	+0.00
	DynaMath-Average $\uparrow$	reasoning	≥ 2.0	42.26	—	—	—	—	—	—	—	—	—	—	—	—	42.36	+0.10
	DynaMath-Worst (NEW) $\uparrow$	reasoning	≥ 6.2	12.77	—	—	—	—	—	—	—	—	—	—	—	—	13.37	+0.60

Per-benchmark noise floor & references

How big a score change has to be before it is real, per benchmark. Both numbers come from the binomial 95% normal approximation (worst case $p=0.5$) on each benchmark's item count n . **Noise $\pm pp$ (single score)** = 95% wobble on one measured accuracy ($1.96 \cdot \sqrt{(p(1-p)/n)}$). **Real Δpp (config change)** = smallest gap between two configs that beats noise ($1.96 \cdot \sqrt{(2p(1-p)/n)}$); it is $\approx \sqrt{2}$ larger because a difference stacks the noise of two scores. These are conservative (independent-sample); same-item paired comparison (McNemar) is somewhat tighter. Published run-to-run variance (DynaMath: $\sim 1\text{--}2pp$ over 5 repeats; multi-run VLM studies: $\sim 1.7\text{--}4.2pp$ std; GPT-judge extraction adds noise for MathVista/MathVerse) sits at or below these floors. Reference column = the dataset paper (which fixes n) plus any measured variance source. MathVerse-Overall averages 5 splits, so its effective floor is somewhat below the per-split value shown.

Benchmark	n	Noise $\pm pp$ (single score, 95%)	Real Δpp (config change, 95%)	Reference (defines n; + variance source)
MathVista-Overall $\uparrow$	1000	± 3.1	4.4	MathVista, arXiv:2310.02255
MathVista-General (NEW) $\uparrow$	460	± 4.6	6.5	MathVista, arXiv:2310.02255
MathVista-Math (NEW) $\uparrow$	540	± 4.2	6.0	MathVista, arXiv:2310.02255
MathVerse-Overall $\uparrow$	788*	± 3.5	4.9	MathVerse, arXiv:2403.14624
MathVerse-T-D $\uparrow$	788	± 3.5	4.9	MathVerse, arXiv:2403.14624
MathVerse-T-L $\uparrow$	788	± 3.5	4.9	MathVerse, arXiv:2403.14624
MathVerse-V-I $\uparrow$	788	± 3.5	4.9	MathVerse, arXiv:2403.14624
MathVerse-V-D $\uparrow$	788	± 3.5	4.9	MathVerse, arXiv:2403.14624
MathVerse-V-O $\uparrow$	788	± 3.5	4.9	MathVerse, arXiv:2403.14624
MathVision $\uparrow$	304	± 5.6	7.9	MATH-Vision, arXiv:2402.14804
MMStar-Overall $\uparrow$	1500	± 2.5	3.6	MMStar, arXiv:2403.20330
MMStar-Math $\uparrow$	250	± 6.2	8.8	MMStar, arXiv:2403.20330
DynaMath-Average $\uparrow$	5010	± 1.4	2.0	DynaMath, arXiv:2411.00836 (rep. var. $\sim 1\text{--}2pp$)
DynaMath-Worst (NEW) $\uparrow$	501	± 4.4	6.2	DynaMath, arXiv:2411.00836 (rep. var. $\sim 1\text{--}2pp$)

Previously-blocked models — now resolved

All four models that completed the 9-model enrichment-target set previously had no step-25 score CSVs. As of this update **every one now has on-disk eval data**, so their blockers are resolved. CSV counts are score/acc files under results/25-merge/<model>/ at update time.

Model	Auto-routed math LLM	Prior blocker	Current status
LLaVA-OneVision-7B	Qwen2-Math-7B (HF)	VLMEvalKit_brv venv missing the <i>llava</i> Python package; eval jobs died with No module named 'llava' (also hit LLaVA-1.5-7B).	Resolved. 24 merges, 402 score/acc CSVs on disk. (Handoff: notes/vlmevalkit_handoff_llava_next_install.md.)
Qwen2.5-VL-7B	Qwen2.5-Math-7B (HF)	(1) hf-xet Rust panic mid-download of Qwen2.5-Math-7B; (2) eval KeyError: 'Qwen2.5-VL-7B-Instruct' (missing from VLMEvalKit supported_VLM).	Resolved. 24 merges, 357 score/acc CSVs on disk. (Handoff: notes/vlmevalkit_handoff_qwen25vl_7b_registry.md.)
InternVL2.5-8B	internlm2-math-plus-7b (HF)	Merge once killed by TERM_THREADLIMIT + TypeError: dtype (both fixed); eval failed with 'InternLM2ForCausalLM' object has no attribute 'generate' (GenerationMixin not inherited).	Resolved. 24 merges, 449 score/acc CSVs on disk. (Handoff: notes/internvl_handoff_generation_mixin.md.)
Qwen2.5-VL-3B	Qwen2.5-3B-DAPO-math-reasoning	Hard-blocked (run_pipeline.sh exit 1): official Qwen2.5-Math ships only 1.5B/7B (28L), shape-incompatible with the 36-layer Qwen2.5-3B backbone, so no math donor was wired in.	Resolved. Donor jaygala24/Qwen2.5-3B-DAPO-math-reasoning (36-layer, shape-compatible) added to step 25; 4 merges, 74 score/acc CSVs on disk.

Methods note — HallusionBench evaluation

HallusionBench is evaluated with **VLMEvalKit** (image_yorn), reporting the standard triplet of metrics — **aAcc** (per-question accuracy), **fAcc** (per-figure: all questions on a figure correct), and **qAcc** (per-question-pair: the original and the edited/control question both correct) — overall and per VS/VD category. Yes/No answers are obtained by **GPT-assisted extraction**: rule-based matching first, with a GPT judge (gpt-4o-mini) resolving the answers that rule-matching leaves ambiguous. This matches the protocol of the original HallusionBench paper (Guan et al., CVPR 2024) and the OpenVLM leaderboard, which also use GPT-assisted yes/no extraction with the same aAcc/fAcc/qAcc metrics. (We switched from VLMEvalKit's rule-only *exact_matching* fallback to GPT-assisted extraction so our numbers are comparable to those works.)

References

- Guan et al., *HallusionBench: An Advanced Diagnostic Suite for Entangled Language Hallucination & Visual Illusion in Large Vision-Language Models*, CVPR 2024. [arXiv:2310.14566](#) (defines aAcc/fAcc/qAcc and GPT-4 yes/no extraction).
- Duan et al., *VLMEvalKit: An Open-Source Toolkit for Evaluating Large Multi-Modality Models*, ACM MM 2024. [arXiv:2407.11691](#) (the evaluation framework used here; GPT-assisted vs exact-matching extraction). Powers the [OpenVLM leaderboard](#).